

Air Quality and Meteorological Data Analysis Report for Kanpur

1. Introduction

This project focuses on analyzing air quality patterns in Kanpur, India, with an emphasis on understanding pollutant trends and their relationship with meteorological factors. The study utilizes publicly available air pollution datasets from multiple monitoring stations and combines them with meteorological data to derive actionable insights.

Note: The air quality check is done specifically in industrial areas of Kanpur.

2. Objective

The primary objectives of this project are:

- To clean and preprocess air quality and meteorological datasets.
- To explore temporal trends in key pollutants (PM2.5, NO2, SO2, etc.).
- To perform weekday vs weekend comparative analysis.
- To assess the impact of meteorological factors (wind speed, wind direction, temperature, etc.) on pollutant concentrations.

3. Data Description

3.1 Air Pollution Data

- Source: Four monitoring stations in Kanpur: IIT Kanpur, FTI Kidwai Nagar, NSI Kalyanpur, and Nehru Nagar.
- Time Period: Pollution data covers 2010 to 2023 (specific months for some stations).
- Key Parameters: PM2.5, PM10, NO, NO2, NOx, SO2, CO, Ozone, NH3, and other pollutants.

3.2 Meteorological Data

- Source: Hourly data from the same stations.
- Time Period: Primarily 2023 and 2024.
- Parameters: Wind Speed, Wind Direction, Ambient Temperature, Relative Humidity, Barometric Pressure.

4. Methodology

4.1 Data Cleaning & Preprocessing

- Merged station-wise pollutant data into a unified dataset.
- Standardized date formats and removed inconsistencies.

- Resampled data to weekly averages to reduce noise and align with meteorological data.
- Cleaned and combined meteorological datasets per station.
- Merged pollution and meteorological data based on station and date for overlapping periods.

4.2 Exploratory Data Analysis (EDA)

- Plotted weekly, monthly, and seasonal pollutant trends.
- Generated boxplots to observe pollutant distribution.
- Assessed seasonal effects (Winter, Summer, Monsoon, Post-Monsoon) on pollution levels.

4.3 Weekday vs Weekend Analysis

- Classified each day as weekday or weekend.
- Performed independent t-tests to statistically compare pollutant levels.
- Visualized results with boxplots.

4.4 Correlation Analysis with Meteorological Factors

- Created correlation heatmaps to assess relationships between pollutants and weather conditions.
- Plotted PM2.5 and Wind Speed trends per station.
- Investigated potential of wind to influence pollutant dispersion.

5. Key Findings

5.1 Temporal Trends

- PM2.5 and other pollutants exhibit seasonal spikes, particularly during winter months.
- Summer and monsoon months show relatively lower pollution levels.

5.2 Weekday vs Weekend Patterns

- T-tests revealed statistically significant differences for certain pollutants, indicating possible influence of human activities like traffic and industrial operations.
- Boxplots visually confirmed variations in pollutant concentrations between weekdays and weekends.

5.3 Impact of Meteorological Factors

- Correlation analysis indicates a negative relationship between wind speed and PM2.5 levels.
- Higher wind speeds generally correspond to lower pollutant concentrations, confirming wind's role in dispersing air pollutants.
- Temperature and humidity also show moderate influence on pollutant behavior.

6. Practical Implications

- Authorities can use these insights to plan targeted pollution control measures, especially during low wind or winter periods.
- Weekend policies (e.g., traffic restrictions) may positively impact air quality.
- Meteorological conditions must be factored into pollution forecasting and management strategies.

7. Conclusion

This project demonstrates a comprehensive approach to analyzing air quality data and understanding the interplay between pollution and meteorological conditions in Kanpur. The methodologies applied ensure reliable insights that can aid both policymakers and researchers in developing effective air quality management plans.

8. Recommendations for Future Work

- Incorporate more recent and granular daily data for better resolution.
- Extend meteorological data to cover a broader historical period.
- Explore advanced machine learning models for pollution prediction.
- Investigate pollutant source identification using wind direction and spike analysis.

Prepared as part of the Air Quality Analysis Project for Kanpur